# Practical Lab 01 — Setting Up Windows for AI Infrastructure Engineering

**Objective:** Inspect the CPU architecture, logical processor availability, and memory resources exposed to Ubuntu through WSL2.

**Commands Executed**

lscpu

nproc

free -h

You do not need to replace Windows or install Linux as a separate operating system. However, I recommend installing WSL2 (Windows Subsystem for Linux) with Ubuntu for our upcoming AI Infrastructure Engineering labs.

WSL2 allows you to run a Linux environment directly on your Windows computer without dual-booting or installing a traditional virtual machine manually.

**Step 1 — Understand what we need**

| Tool | Purpose | Required now? |
|---|---|---|
| Windows PowerShell | Inspect your Windows system | Yes |
| WSL2 | Run Linux on Windows | Recommended |
| Ubuntu | Linux distribution for commands and engineering labs | Recommended |
| Windows Terminal | Manage PowerShell and Ubuntu terminals | Optional |
| VS Code | Edit scripts and GitHub documentation | Later |
| Docker | Run containerized applications | Not yet |
| NVIDIA CUDA | Run supported applications on NVIDIA GPUs | No |

**Step 2 — Check your CPU and RAM in Windows**

You can do this immediately without installing anything.

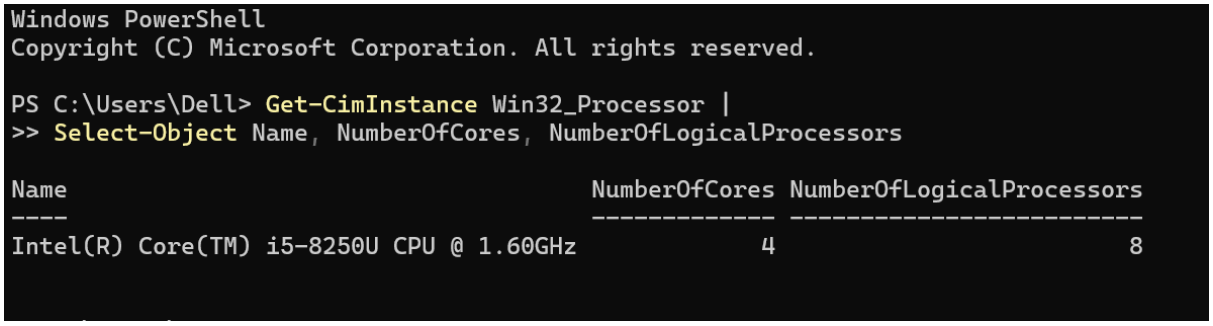
1. Press the Windows key, type PowerShell, and open Windows PowerShell.

2. Copy and execute this command:

```
Get-CimInstance Win32_Processor |

Select-Object Name, NumberOfCores, NumberOfLogicalProcessors
```

This displays your CPU model, physical core count, and logical processor count.




**1. Brand and Model**

- **Intel Core i5-8250U:** This is the specific model of processor. The **i5** represents Intel's mid-range tier (great for everyday tasks and multitasking). The **8** indicates it is an 8th-generation chip, and the **U** means it is an ultra-low power processor designed to save battery life in laptops.

**2. Clock Speed**

- **CPU @ 1.60 GHz:** This is the base clock speed. It means the processor can execute **1.6 billion computing cycles per second** when handling normal tasks. When you do something heavy (like opening a large app or editing a video), this chip can automatically speed up (turbo boost) past this number.

**3. Physical Cores**

- **No. of cores: 4:** Cores are the actual, physical "brains" inside the processor chip. Having 4 physical cores means your computer has four separate processing units that can physically cut up and work on different tasks simultaneously.

**4. Logical Processors (Threads)**

- **No. of logical processors: 8:** These are also known as threads. Your processor uses a technology called **Hyper-Threading**, which allows a single physical core to handle two tasks at the exact same time. To your operating system (Windows/macOS), it looks and behaves as if you have **8 separate processors** working together, making multitasking much smoother.

## What are the benefits of cores and threads in CPU or GPU

Cores and threads are the processing engines of a CPU or GPU, working together to determine how many tasks your computer can handle simultaneously and how quickly it can finish them.

While cores represent the physical hardware engines, threads act as the virtual lanes feeding data into those engines.

### Benefits of Cores and Threads in a CPU

A CPU is designed for **sequential processing**—handling complex, varied tasks one after another very quickly (like running an operating system, loading a web page, or calculating game logic).

- **Benefits of More Cores (Physical Brains):**

    - **True Hardware Multitasking:** Each physical core can process an entirely independent program. With 4 or 8 cores, you can stream music, download a file, run a virus scan, and type a document at the exact same time without a single stutter.
    - **Heavy Workload Distribution:** Demanding applications like video editing software (Premiere Pro) or 3D rendering engines can split a massive project into chunks, giving one chunk to each core to finish the job exponentially faster.
- **Benefits of More Threads (Virtual Lanes / Hyper-Threading):**
    - **Zero Idle Time:** A single CPU core is so fast that it often wastes time waiting for data to arrive from your RAM. Hyper-Threading creates two threads (logical processors) per core, allowing the core to instantly switch to task B if task A is delayed.
    - **Smoother System Responsiveness:** More threads prevent your computer from freezing up or lagging when background system tasks unexpectedly spike.

**Benefits of Cores and Threads in a GPU**

A GPU is fundamentally different from a CPU. It is built for **massive parallel processing**—handling thousands of tiny, repetitive math calculations all at the exact same time (like calculating the color and lighting of millions of individual pixels on your monitor).

- **Benefits of Thousands of GPU Cores:**
    - **Instant Image and Frame Rendering:** Instead of having 4 or 8 powerful cores like a CPU, a GPU has *thousands* of smaller, specialized cores (e.g., NVIDIA CUDA cores or AMD Stream processors). This allows the GPU to calculate the physics, textures, and shading of an entire 3D video game frame at 60 to 144+ frames per second.
    - **Accelerated AI and Machine Learning:** Modern AI workloads rely on massive matrix multiplication. A high core-count GPU can train AI models or process machine learning algorithms in minutes, whereas a CPU would take days.
- **Benefits of Hardware GPU Threads:**
    - **Massive Thread Concurrency:** GPUs do not use hyper-threading in the way CPUs do. Instead, they launch **thousands of threads simultaneously** across their massive core arrays. If a few hundred threads are waiting on data, the GPU scheduler instantly swaps in a few hundred other threads, hiding latency completely and keeping the hardware saturated.

**3. Next, check your installed RAM:**

```
Get-CimInstance Win32_ComputerSystem |

Select-Object
@{Name="TotalRAM_GB";Expression={[math]::Round($_.TotalPhysicalMemory/1GB,2)}}
```

```
PS C:\Users\Dell> Get-CimInstance Win32_ComputerSystem |
>> Select-Object @{Name="TotalRAM_GB";Expression={[math]::Round($_.TotalPhysicalMemory/1GB,2)}}

TotalRAM_GB
-----------
      15.84
```

**4. Check your Windows version:**

winver

A small window will open showing your Windows edition, version, and OS build.




**Windows 11 version 25H2 (OS Build 26200.9547)** refers to a specific, fully updated production release of the Windows 11 2025 Update. [1, 2]

**What This Build Breakdown Means**

- **25H2 (Version):** This indicates it is the major feature update released for the **second half of 2025**.
- **26200 (Major Build):** This is the core operating system build code family specifically designated for the **25H2 release**.
- **.9547 (Revision Number):** This specific minor revision represents a system updated with cumulative security patches and hotfixes released in **September/October 2026**.

## What is WSL?

WSL stands for Windows Subsystem for Linux. It is a feature built into Windows that allows you to **run a full, native Linux environment directly on your Windows computer** without the heavy overhead of a traditional virtual machine or the need to dual-boot your PC.

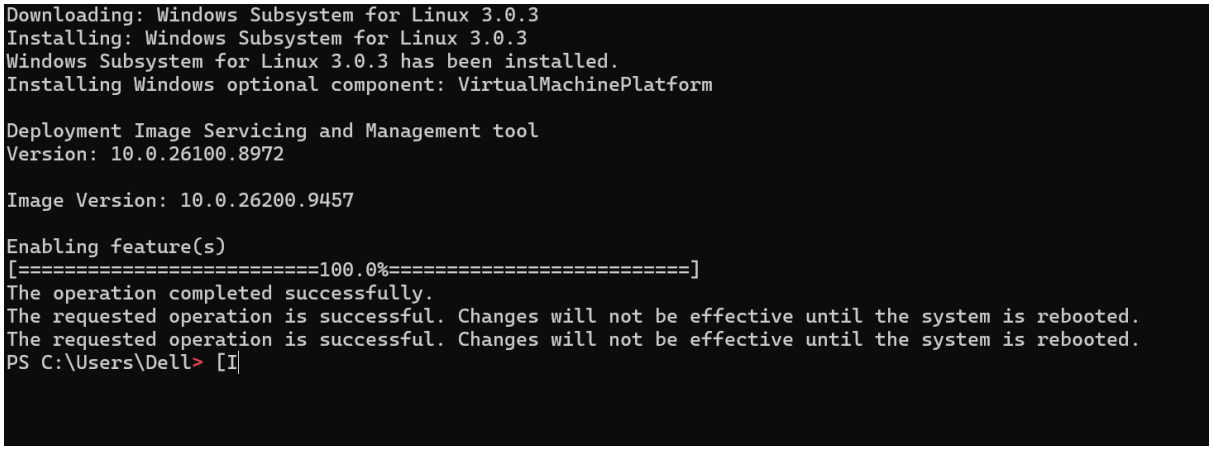
With WSL, you can open a terminal and run Linux distributions (like Ubuntu, Debian, or Fedora) right alongside your normal Windows applications.

**5. Check whether WSL is already installed:**

wsl --status

If WSL is not installed, Windows may show a message indicating that it is unavailable or needs installation. That is expected.

The Windows Subsystem for Linux is not Installed. You can install by running ‘wsl.exe –install’.

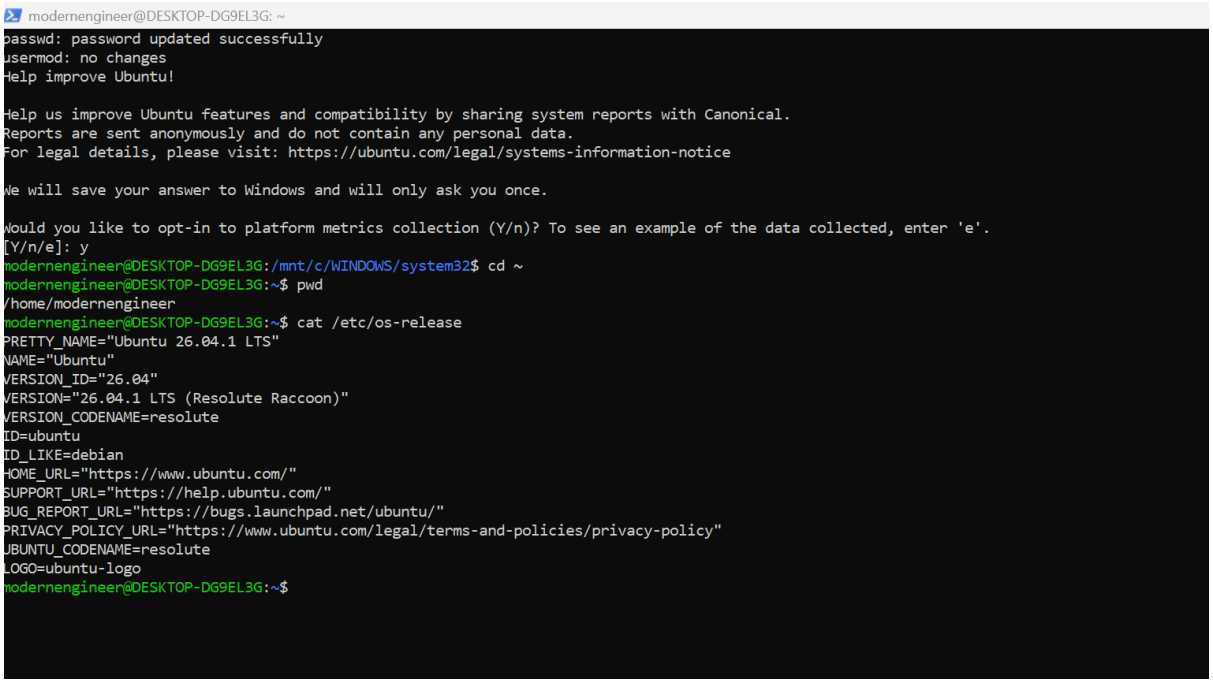



Restart you windows — Using PowerShell
Run this command:
shutdown /r /t 0

**Step 3 — Install Ubuntu using WSL2**
Once you have confirmed that you're running a compatible Windows version and have sufficient available memory and disk space, follow these steps.
1. Search for PowerShell in Windows.
2. Right-click Windows PowerShell → Run as Administrator.
3. Execute:
wsl --install -d Ubuntu

4. Restart Windows if prompted.
5. Open Ubuntu from the Windows Start menu.
The first launch may ask you to create a Linux username and password. This password is separate from your Windows password. The terminal will not display characters while you type it.
6. Verify the installation from PowerShell:
wsl -l -v
Look for Ubuntu with VERSION 2

**Step 4 — Run your first Linux practical lab**
Open Ubuntu, not PowerShell.
First, verify the operating system:
cat /etc/os-release
You should see Ubuntu information.
Then execute the commands individually.




### Step 3 — Start CPU Practical Lab 01

Now run:

lscpu

This is our first actual AI Infrastructure hardware inspection command.

It displays important CPU information, including processor architecture, logical processors, cores, threads, and CPU topology.

### Practical Lab 01 — CPU Architecture Analysis

`lscpu` output confirms that Ubuntu is running successfully under WSL2. We can now examine your actual processor specifications and relate them to AI Infrastructure Engineering.

#### 1. CPU specifications

| Parameter | Your System |
|---|---|
| CPU Model | Intel Core i5-8250U |
| CPU Architecture | x86_64 |
| CPU Family | 6 |
| Logical CPUs visible | 8 |
| Threads per Core | 2 |

| | |
|---|---|
| Cores per Socket | 4 |
| Sockets | 1 |
| L1d Cache | 128 KiB (4 instances) |
| L1i Cache | 128 KiB (4 instances) |
| L2 Cache | 1 MiB (4 instances) |
| L3 Cache | 6 MiB (1 instance) |
| Hypervisor | Microsoft |
| Virtualization Type | Full |
| NUMA Nodes | 1 |

Important: These are the CPU resources and topology reported to Ubuntu through WSL2. The underlying processor is an Intel Core i5-8250U, with four physical cores and eight hardware threads.

**2. Understanding CPU results**

**A. CPU cores vs logical CPUs**

System reports:

CPU(s): 8

Thread(s) per core: 2

Core(s) per socket: 4

Socket(s): 1

This means processor has 4 physical cores and 8 logical processors.

Intel Hyper-Threading allows each physical core to expose two logical execution contexts, which can help improve utilization of the processor's execution resources.

However, eight logical processors do not provide the same computing capacity as eight independent physical cores.

**B. CPU architecture — x86_64**

CPU uses the x86_64 instruction set architecture, also called AMD64.

Most commonly used Linux distributions, Docker images, and development tools support this architecture.

This will simplify compatibility for our early labs.

**C. CPU cache**

CPU reports L1, L2, and L3 cache levels.

These caches reduce the need for the processor to repeatedly access comparatively slower system RAM.

They are particularly relevant to workloads with frequent data access, including CPU-based inference and data preprocessing.

**D. NUMA architecture**

output reports:

NUMA node(s): 1

This means Ubuntu currently sees one NUMA node.

Later, when studying multi-GPU servers and AI clusters, we will revisit CPU sockets, memory locality, PCIe topology, and NUMA-aware workload placement.

**E. Microsoft hypervisor**

Hypervisor vendor: Microsoft

Virtualization type: full

This is expected because WSL2 runs Linux using lightweight virtualization.

It does not mean Ubuntu has replaced Windows. Both environments can coexist.

## Observations

The processor contains four physical cores and exposes eight logical processors through Intel Hyper-Threading.

Ubuntu is running in a WSL2 virtualized environment using the Microsoft hypervisor.

The Linux environment reports approximately 7.7 GiB of memory capacity, with approximately 7.2 GiB available at the time of inspection.

The system is suitable for Linux administration, system monitoring, lightweight containerization, networking experiments, and CPU-based AI workloads.

GPU-dependent CUDA and NVIDIA infrastructure labs will require access to compatible NVIDIA hardware.

## Engineering Takeaways

1. Physical CPU cores and logical processors are not equivalent.
2. WSL2 allows Linux infrastructure tools to run on Windows.
3. Available memory is more useful than completely unused memory when assessing capacity for additional workloads.
4. CPU and memory inspection are basic infrastructure capacity-assessment techniques.

5. Local hardware limitations must be considered when designing AI infrastructure experiments.